Mariam Essam

AI Engineer
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SUMMARY

Senior Computer Engineering student with strong knowledge in ML, computer vision, NLP, and LLMs. Experienced in building multi-agent AI systems and RAG pipelines using LangChain, LangGraph, LlamaIndex, and ChromaDB, with hands-on expertise in prompt engineering.

EXPERIENCE

Freelancer AI Engineer — 8/2025 - 10/2025

MGD Labs

- Developed a secure digital slide viewer to facilitate remote pathology diagnosis using Python Flask and JavaScript.
- Integrated an AI model for pathology slide analysis to assist with diagnostic verification.

ACHIEVEMENTS

Solved **300+** coding challenges on LeetCode and Codeforces, applying data structures (arrays, trees, graphs, hash maps) and algorithms (DP, recursion, sorting, searching) to solve complex problems.

EDUCATION

2022-2027

Bachelor of Computer Engineering
Helwan University
Grade: Excellent (88%)

PROJECTS

Hieroglyphics Translator Experience

- An optimized computer vision pipeline for real-time hieroglyphic detection and translation, extended into a VR application and MCP server for AI agent integration.
- Model: Custom YOLOv8-based object detection fine-tuned on hieroglyphic datasets
- Backend: MCP (Model Context Protocol) server.
- Frontend: Gradio interface (web).
- Deployment: Hugging Face Spaces, cloud-hosted MCP server.

Underwater Object Detection for Autonomous Vehicle (MATE ROV Competition)

- Fine-tuned YOLOv11 Large for real-time underwater trash and crab detection on an autonomous ROV, applying extensive data augmentation to simulate underwater conditions and maximize detection accuracy **— developed for the MatROV Competition**
- Built a 3D modeling system for underwater objects using ROV camera footage as input, enabling 360° object reconstruction from rotating video sequences

Facial Key points Detection

- Built and trained a facial landmark detection system from scratch without relying on pre-trained models or transfer learning.
- Designed custom image augmentation functions to improve generalization.

Image Caption Generator

- Developed an Encoder-Decoder model for multi-object recognition across 100+ object classes, generating structured captions capturing attributes and spatial relationships.

RAG-based Research Paper Assistant

- Built a RAG system for research paper Q&A by chunking and embedding documents, storing vectors in ChromaDB for fast semantic retrieval, and integrating an LLM for context-grounded answer generation.

Real-time speech-to-speech model

- Developed an end-to-end system that converts spoken input into text, analyzes the content using LLMs, and generates natural-sounding synthesized responses. The model maintains speaker identity and emotional tone during conversion.

Breast Cancer Detection from Mammograms

- Used the INbreast and CBIS-DDSM datasets, integrating DICOM image metadata.
- Handled metadata extraction and mapped labels and images for accurate training and evaluation.
- Implemented a convolutional neural network (CNN) for binary classification using PyTorch.
- Plotted BI-RADS distribution and explored tumor patterns for insights into dataset characteristics.

NLP Medical Report Simplification

- Built a web application to simplify complex medical reports into patient-friendly language.
- Uses OCR to extract text from uploaded images. The OCR process handles different fonts, layouts, and image qualities.
- Designed a user-friendly interface with Streamlit for ease of use.

Automation Projects

- Used Selenium to automate web interactions and developed Instagram and facebook automation scripts.

EXTRACURRICULAR ACTIVITIES

Founder and Leader, Data Analysis Student Activity

- Created a student activity that aims to teach students from different backgrounds data analysis and how to use it in their own field of study.
- Created content for classes including on-site and online lectures, as well as marketing content for social media presence.

SKILLS

- **Languages**: Python, C++
- **Computer Vision:** OpenCV, YOLO
- **LLMs & GenAI**: LangChain, LangGraph, LlamaIndex, RAG, Prompt Engineering, ChromaDB
- **Frameworks**: PyTorch, TensorFlow

LANGUAGE

- **Arabic**: Native
- **English**: Native-level proficiency